In the name of Allah

Signals and Systems
Computer Homework 3

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Section 1

Part a

In my code, I first found the zeros and poles of $H_1(s)$ and $H_2(s)$ and checked whether the real parts of all poles are strictly less than zero. The results were then displayed in the terminal.

```
>> SAS_CHW3_Q1
system H1 is stable
system H2 isn't stable
```

Figure 1: Terminal output showing system stability results.

Additionally, I plotted the pole-zero diagrams for both $H_1(s)$ and $H_2(s)$.

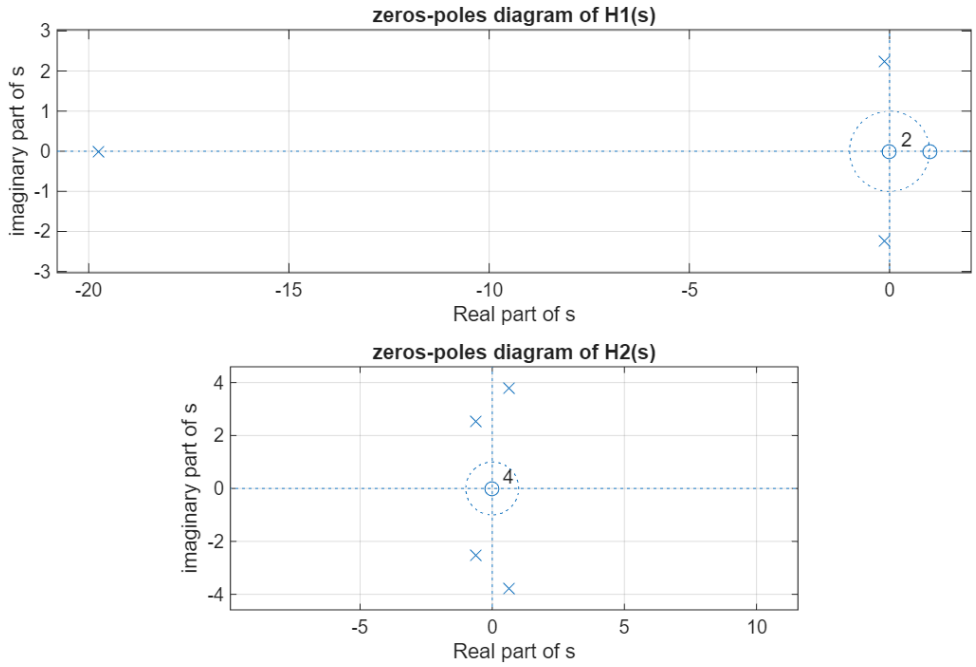


Figure 2: Pole-zero diagrams of $H_1(s)$ and $H_2(s)$.

Part b

It is worth noting that the `pzmap` function plots the zeros and poles, but not in the classic complex plane format. Therefore, we utilized grid lines to identify their locations more precisely.

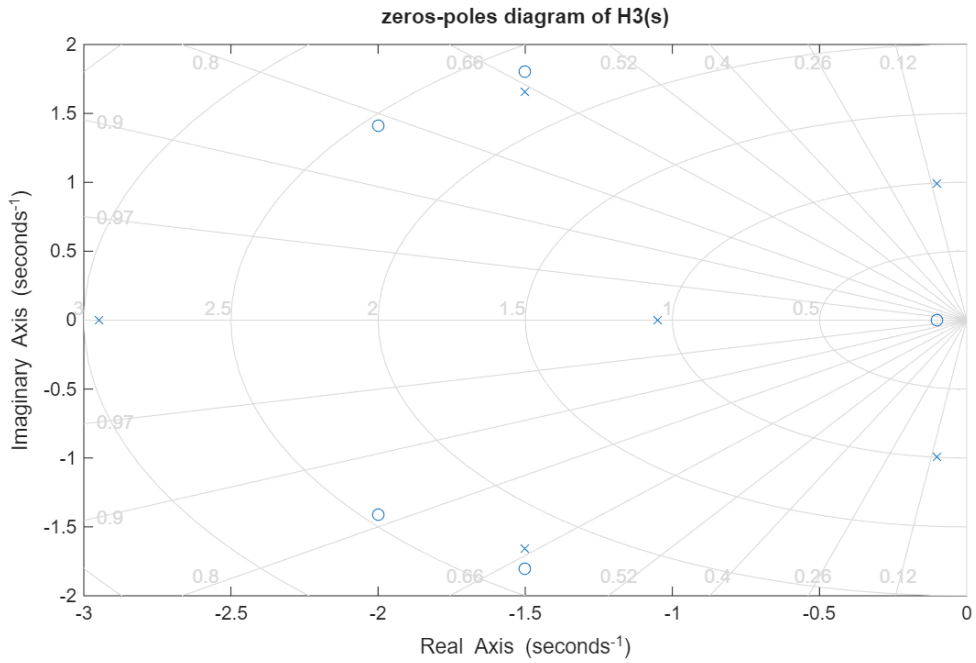


Figure 3: Step responses of systems $H_1(s)$ and $H_2(s)$.

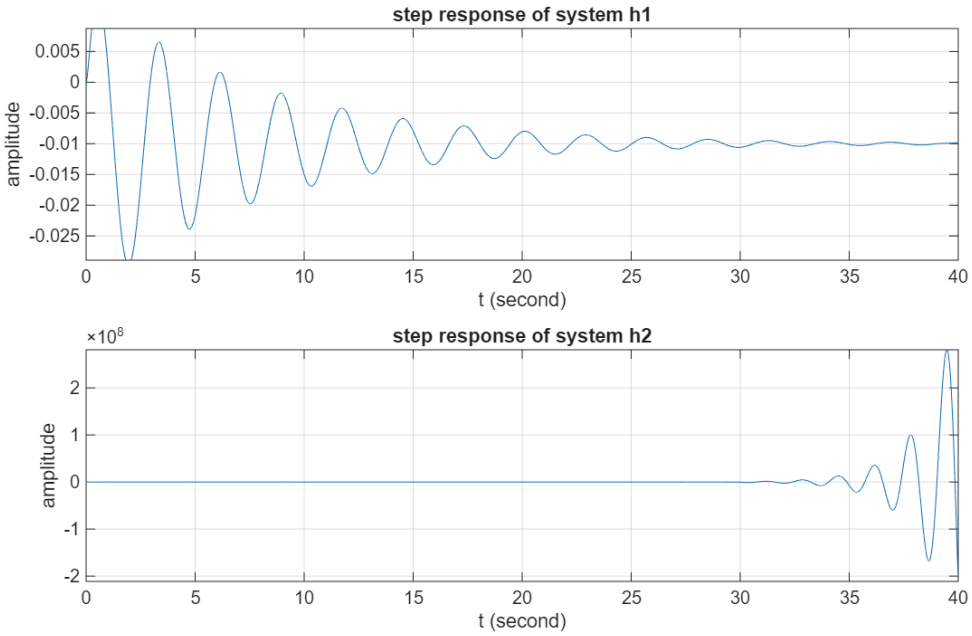


Figure 4: Pole-zero diagram of $H_3(s)$.

Part c

An LTI system with an impulse response $h(t)$ is bounded-input bounded-output (BIBO) stable if and only if for every bounded input $x(t) \in L_\infty$, the output is also bounded, i.e., $y(t) \in L_\infty$:

$$h(t) \text{ is stable} \iff \forall x(t) \in L_\infty : y(t) \in L_\infty$$

Since the unit step function satisfies $u(t) \in L_\infty(\mathbb{R})$, if the step response satisfies $y(t) \notin L_\infty(\mathbb{R})$, the system is unstable:

$$\text{If } y(t) \notin L_\infty(\mathbb{R}) \implies h(t) \text{ is unstable}$$

Based on the observed step responses, $h_1(t)$ appears to be stable, whereas $h_2(t)$ is unstable, which completely aligns with the numerical results obtained in Part (a).

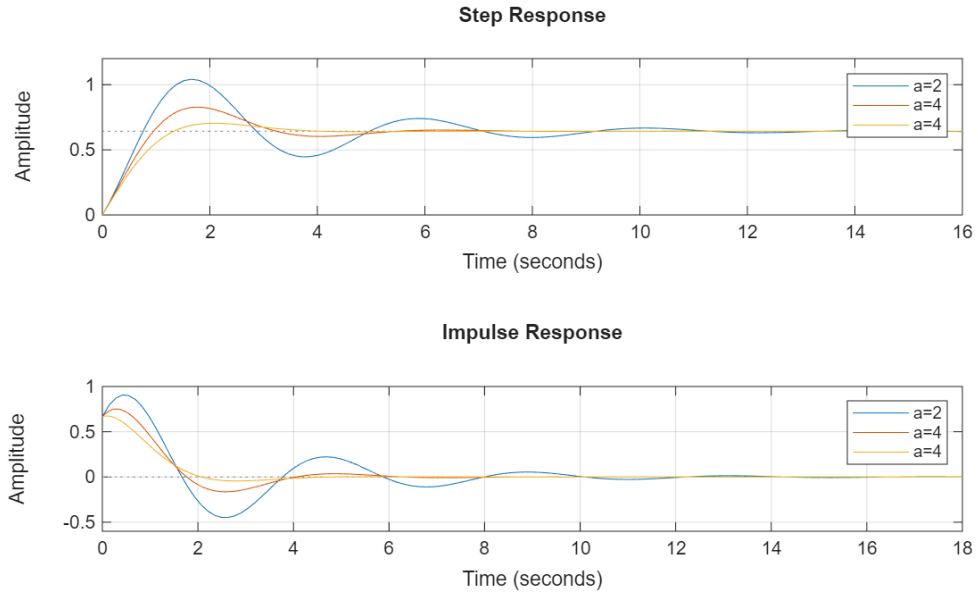


Figure 5: Step and impulse responses for different values of parameter a.

Part d

For all $a > 0$, the real parts of the roots of the characteristic polynomial (denominator) $s^2 + as + 7$ are negative. As a increases, the real parts of these poles become increasingly negative. Consequently, the amplitude of the frequency response $G(j\omega)$ decreases near the imaginary parts of the poles (and everywhere along the $j\omega$ axis), and as a result:

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(j\omega) e^{j\omega t} d\omega$$

$$|g(t)| = \frac{1}{2\pi} \left| \int_{-\infty}^{\infty} G(j\omega) e^{j\omega t} d\omega \right|$$

$$|g(t)| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega) e^{j\omega t}| d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(j\omega)| d\omega$$

Therefore, the magnitude of $g(t)$ decreases as well. This behavior is clearly visible in the diagrams provided above.

Section 2

Part a

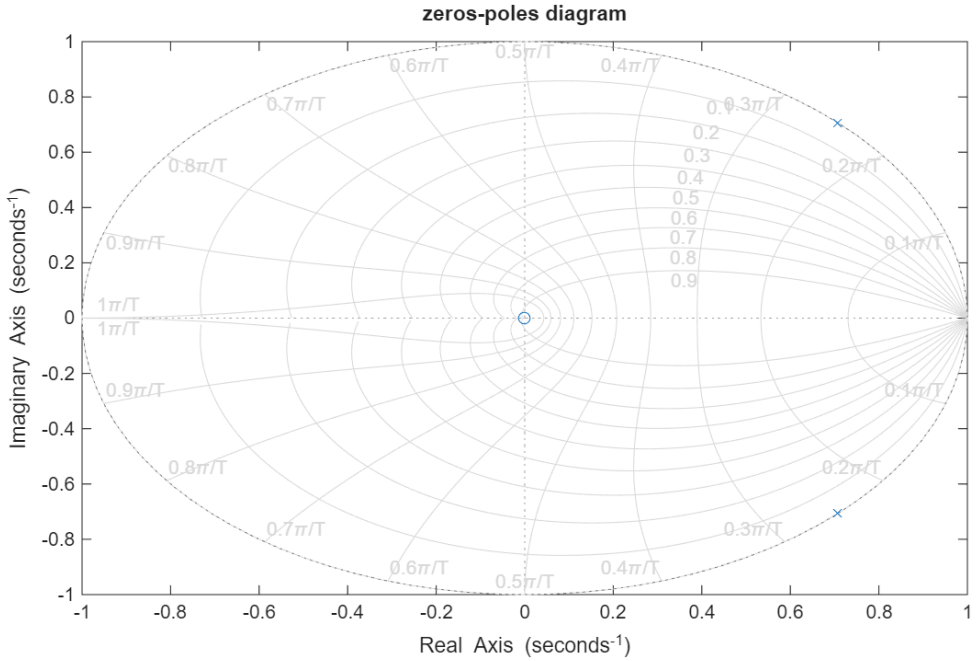


Figure 6: Pole-zero diagram in the Z-plane.

Part b

```
maximum of amplitude of poles is:
     1
```

Figure 7: Terminal output showing the maximum pole amplitude.

The maximum magnitude of the poles is found to be 1. Given that $x[n]$ is a right-sided signal, the Region of Convergence (ROC) must extend outward from the outermost pole. Therefore, the ROC is defined as:

$$\text{ROC} = \{z \mid |z| > 1\}$$

Part c

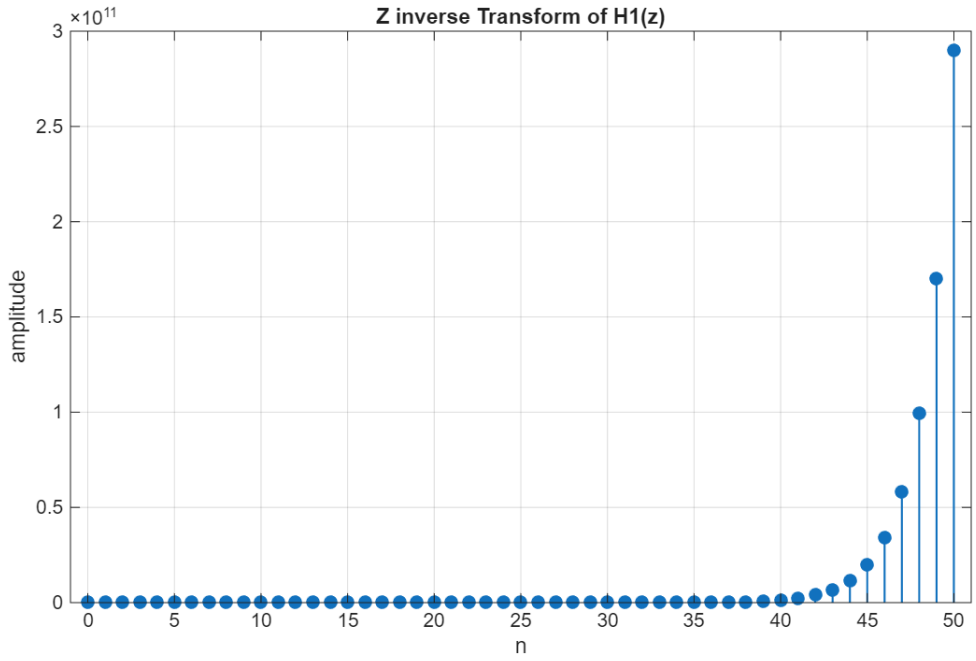


Figure 8: Inverse Z-transform of $H_1(z)$.

Part d

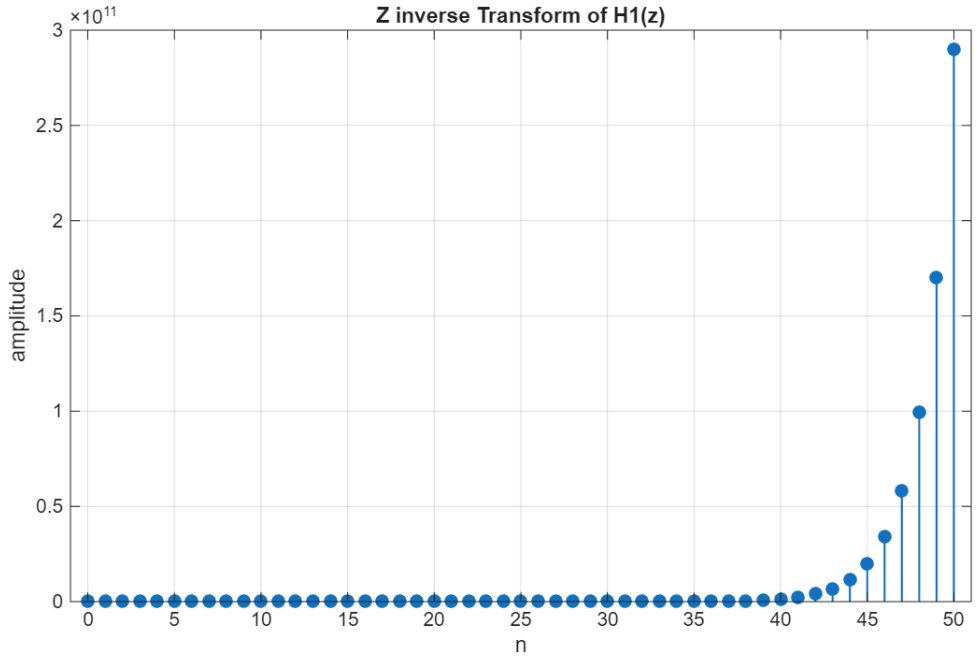


Figure 9: Verification of the Inverse Z-transform of $H_1(z)$.

The two results are exactly the same.

Section 3

Part a

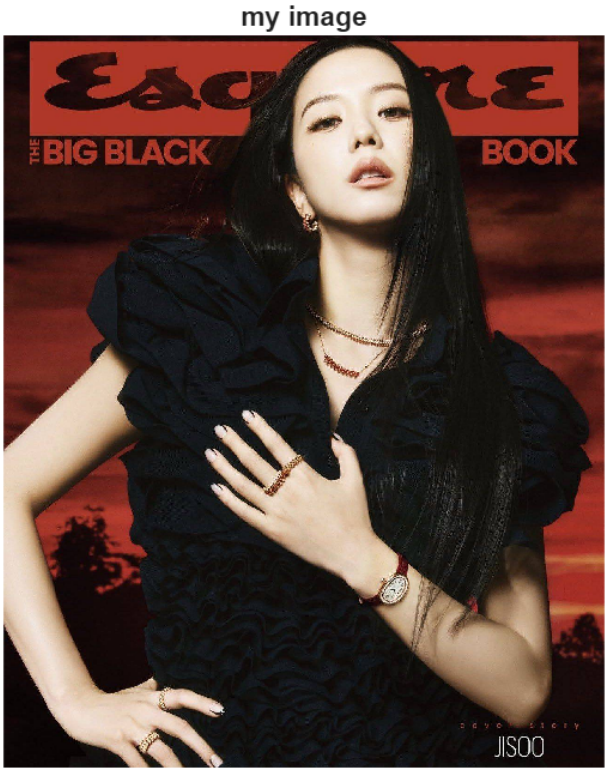


Figure 10: Original input image.

Part b

The brightness of the red channel is significantly higher than that of the other channels. This indicates that the red component plays the dominant role in the image, which is primarily due to the skin tones and the warm background coloration.



Figure 11: Original image alongside its R, G, and B color channels.

Part c

The `rgb2gray` function combines the color channels using a weighted sum rather than a uniform average. Consequently, a close inspection (particularly around the subject's left hand) reveals subtle differences between the results of these two methods.

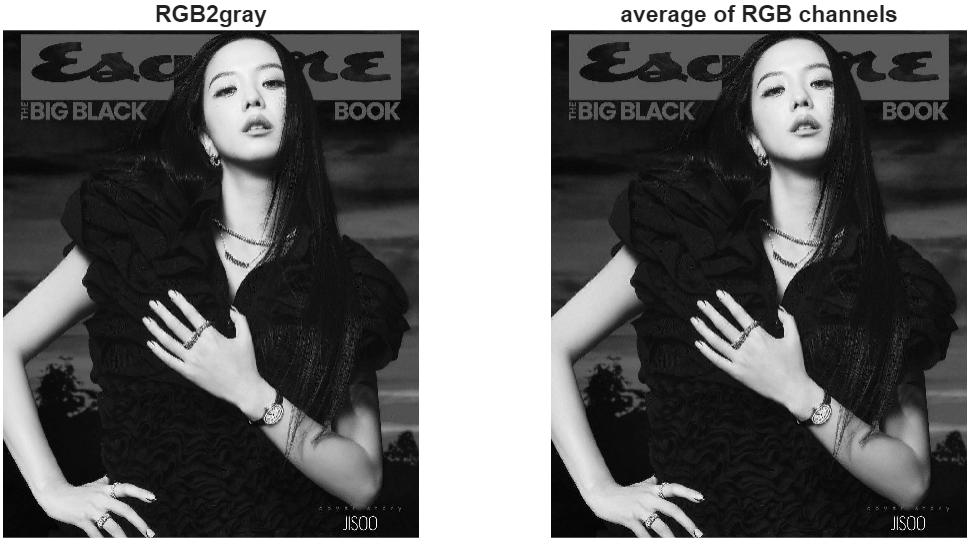


Figure 12: Comparison between `rgb2gray` and uniform averaging.

Part d



Figure 13: High-pass filtering/edge detection applied to the city image.

(if you wondered what would happen for the first part image if I input it to this system):



Figure 14: High-pass filtering/edge detection applied to the original image.

This is a differencing system acting as a high-pass filter. It is highly sensitive to sharp intensity variations or edges, which correspond to high-frequency components. The image of the city features many intricate details and sharp lines, resulting in a high concentration of high frequencies. In contrast, the human subject features smoother transitions, which naturally contain lower frequencies.